

Tri-Temp High-Throughput IC Test Handler

Authoritative Model Technical Manual & Diagnostic Reference | VAI-MAN-TH-001

NOTICE: SYNTHETIC PROTOTYPE TECHNICAL MANUAL

This document is artificially generated for the VectorAI demonstration and software development.

The specifications, thresholds, service-life values, maintenance intervals, and operating parameters are synthetic and must not be used for real industrial equipment operation or maintenance.

1. Machine Overview & Manufacturing Context

Equipment Model:	Tri-Temp High-Throughput IC Test Handler	Document ID:	VAI-MAN-TH-001
Machine Type Key:	test-handler	Cleanroom Bay / Stage:	Bay 6A: Tri-Temp Final Test
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Automated tri-temperature IC test handler for presenting packaged semiconductor devices to Automated Test Equipment (ATE) at hot (+150°C), ambient (+25°C), and cold (-40°C) conditions.

Process Flow: Packaged ICs are loaded into soak chambers where LN2 injection or electric heaters stabilize device temperature. High-speed pick-and-place arms insert devices into Kelvin contact sockets. ATE testers execute parametric and functional tests, and the handler sorts ICs into Pass, Fail, and Speed bins.

2. Major Subsystems & Critical Components

Subsystem / Component	Primary Function	Key Parameters	Degradation Indicators
Kelvin Test Socket Contactor	Provides low-resistance electrical contact to IC pins during testing.	Contact resistance (mΩ), pogo pin force (N), contact cycle count.	Contact resistance spike (>50 mΩ), false failure rejections.
Tri-Temp Thermal Soak Chamber	Stabilizes IC package temperature at -40°C to +150°C.	Chamber temperature (°C), LN2 flow rate, heater output (%).	Temperature oscillation (>±2°C), LN2 valve icing.
Pick-and-Place Theta Arm	Inserts and extracts devices into test socket with high speed.	Arm vibration (mm/s), theta rotation angle, cycle time (sec).	Arm vibration rise (>0.7 mm/s), socket insertion misalignment.

3. Sensor Telemetry & Threshold Matrix

Threshold boundaries configured for real-time telemetry monitoring and automatic anomaly triggers:

Sensor Name & ID	Unit	Normal Range	Warning Range	Critical Range	Direction
Handler Vibration vibration_handler	mm/s	0.1 – 0.55	0.55 – 1.1	1.1 – 2	Higher is Worse
Socket Actuator Load load_actuator	N	40 – 60	60 – 80	80 – 120	Higher is Worse

4. Deterministic RUL Model (Weighted Degradation)

Base Useful Life: 2,100 operating hours. **Model:** $RUL = BaseLife * (1 - (0.50 * wear_handler_vib + 0.30 * wear_actuator + 0.20 * wear_chamber_temp))$

Parameter	Sensor ID	Unit	Weight	Healthy Limit	Critical Limit	Direction
Handler Vibration	vibration_handler	mm/s	0.50	0.5	1.2	HIGHER IS WORSE
Actuator Load	load_actuator	N	0.30	50	85	HIGHER IS WORSE
Chamber Temp	temperature_chamber	°C	0.20	85	110	HIGHER IS WORSE
TOTAL WEIGHT SUM	-	-	1.00 (100%)	-	-	VERIFIED OK

5. Troubleshooting & Anomaly Symptoms Matrix

Symptom & ID	Severity	Related Sensors	Possible Root Causes	Recommended Corrective Action
Parametric Test False-Reject Spike > 1.2% SYM-TH-001	HIGH	load_actuator, vibration_handler	• Kelvin socket pogo pin contamination • Socket actuator force drift • Theta positioning error	Replace pogo pin contactor block and clean optical theta encoder.

6. Authoritative Diagnostic Knowledge Base (Failure Scenarios)

Scenario 1: Kelvin RF Socket Pogo Pin Contact Resistance Elevation (SCEN-TH-001)	Severity: WARNING
Telemetry Sensor Pattern:	load_actuator > 65 N, vibration_handler > 0.70 mm/s
Possible Root Causes:	Pogo pin oxide buildup after 120,000 cycles Actuator guide friction
Recommended Corrective Action:	Replace 128-pin Kelvin high-frequency contactor block and clean optical rotary encoder.
Verification & Recovery Steps:	Verify 4-wire Kelvin contact resistance (< 25 mΩ) > Run 50-device golden unit verification test (100% pass)